Real Estate Investment Analysis

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Business Objective

- Identify good properties to invest in
- Understand factors affecting property prices

Business Problem

People invest without proper analysis, we are analyzing which property is a good investment:

- Undervalued price (cheaper than similar properties nearby)
- High demand drivers (transport, schools, hospitals)
- Future appreciation potential (developing locality, newer property)
- Rental attractiveness (amenities, furnishing, accessibility)

We are analyzing data of properties across India which contains 250000 rows and 22 columns

Core variables:

- **Location**: State, City, Locality
- **Property:** Property_Type, BHK, Size_in_SqFt
- **Price**: Price_in_Rs, Price_per_SqFt

Infrastructure :

- Nearby_Schools, Nearby_Hospitals, Public_Transport_Accessibility, Parking_Space, Security

Property features:

- Year_Built, Age_of_Property, Furnished_Status, Floor_No, Total_Floors

Amenities: Clubhouse, Gym, Pool, Garden, Playground

Data Preprocessing

Handle missing values and duplicates: The data is free from missing and duplicate values

- For easy calculation and understanding changed the below columns
- **Price_in_Rs =** Price_in_lakhs * 100000
- Modifying existing Price_per_SqFt column with new calculation as the value is given in lakhs and rounded off to near value
- **Price_per_SqFt** = Price_in_Rs/ Size_in_SqFt

Deleting columns

- As Total_floors cannot be less than Floor_no we can remove those rows, but they contribute to 116304 rows, so instead replaced those property_type with Independent house and villa with 1 which satisfy the above condition. So we are reducing those to 38,639 rows are deleted.

Adding Columns

- Based on the Amenities column added columns (Garden, Pool, clubhouse, gym, playground and amenities_count) by creating amenities dummies columns
- **Age_of_Property** = current_year - Year_Built
- **Local_avg_price :** find median of Price_per_SqFt by grouping city and locality
- **Undervalued:** Price_per_SqFt < local_avg_price
- **Age_score :** Age_of_Property <= 5 then 3, Age_of_Property <= 15 then 2 else 1
- **Livability_score :** amenities_count + Parking_Space (yes =1, no =0) + Security (yes =1, no =0)
- **Investment_score :** undervalued*3 + location_score*2 + age_score*2 + livability_score
- **Appreciation_rate**: sort values by 'City', 'Locality', 'Year_Built' and then find pct_change of Price_per_SqFt groupby 'City', 'Locality'
- **Price_bucket: "<20L", "20-50L", "50L-1Cr", "1-2Cr", "2-5Cr"**
- **Size_bucket: "<1K", "1K-2K", "2K-3K", "3K-4K", "4K-5K"**

Good Properties:

- Investment score > 75% of investment score

Good Investment :

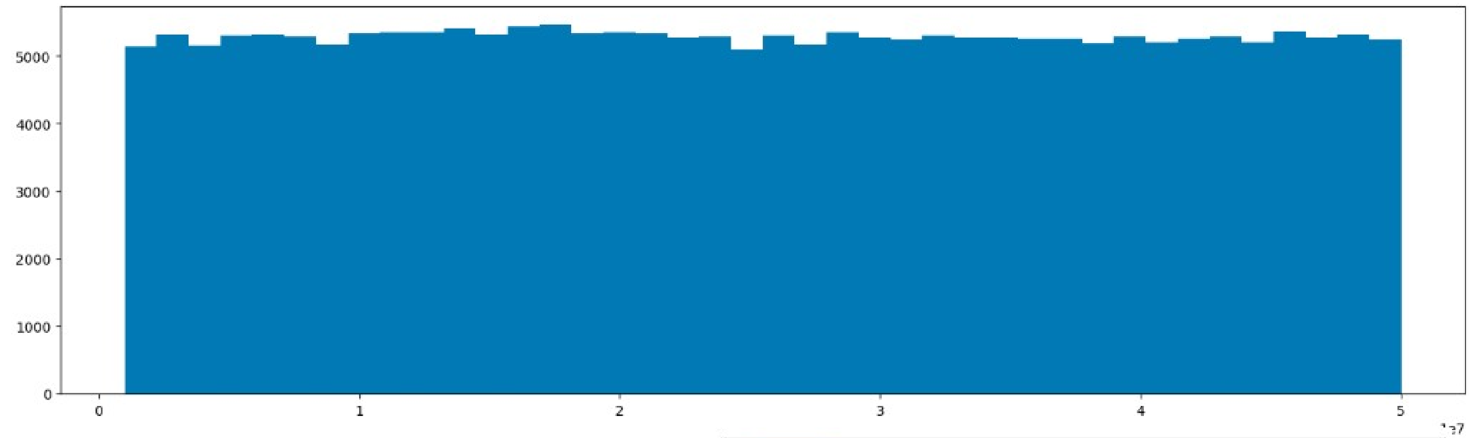
- Appreciation rate > threshold

	BHK	Size_in_SqFt	Price_per_SqFt	Year_Built	Floor_No	Total_Floors	Age_of_Property	Nearby_Schools	Nearby_Hospitals	Price_in_Rs
min	1.0	500.0	202.0	1990.0	0.0	1.0	3.0	1.0	1.0	1000000.0
max	5.0	5000.0	99182.0	2023.0	30.0	30.0	36.0	10.0	10.0	50000000.0

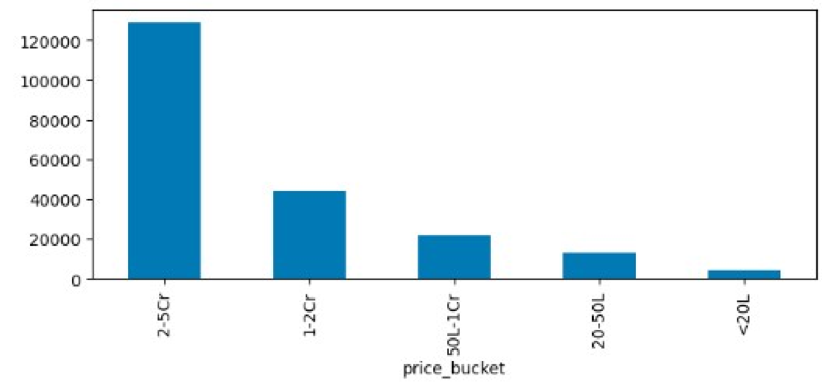
Clubhouse	Garden	Gym	Playground	Pool	amenities_count	local_avg_price	location_score	age_score	livability_score	investment_score	appreciation_rate
0.0	0.0	0.0	0.0	0.0	1.0	782.0	2.0	1.0	1.0	7.0	-99.669834
1.0	1.0	1.0	1.0	1.0	5.0	50173.0	22.0	3.0	7.0	60.0	35463.485477

	State	City	Locality	Property_Type	Furnished_Status	Public_Transport_Accessibility	Parking_Space	Security	Amenities	Facing	Owner_Type	Availability_Status
unique	20	42	500	3	3	3	2	2	325	4	3	2
top	Tamil Nadu	Coimbatore	Locality_296	Villa	Unfurnished	High	No	Yes	Pool	West	Broker	Under_Construction

What is the distribution of property prices?

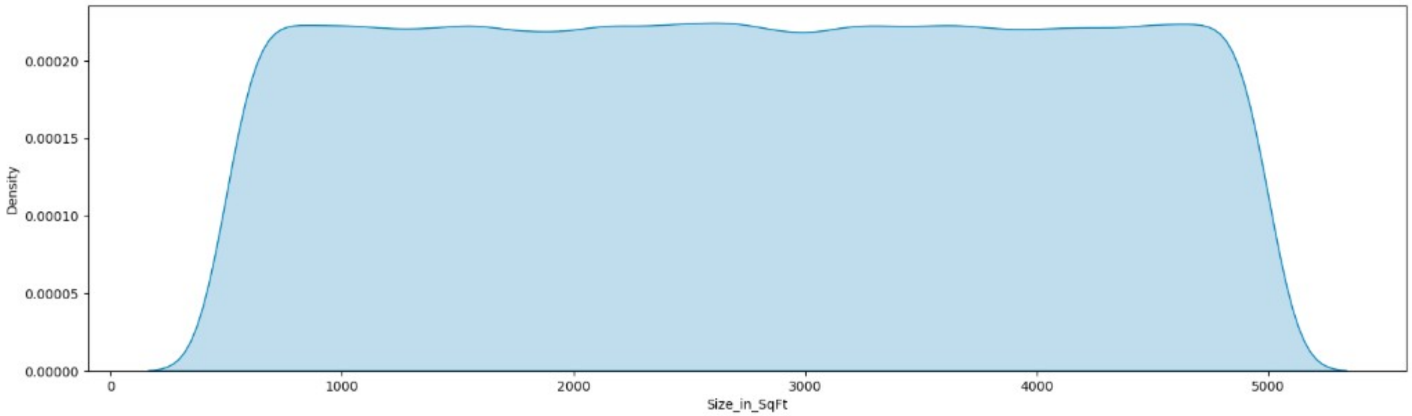



```
price_bucket
2-5Cr       128956
1-2Cr        43844
50L-1Cr      21531
20-50L       12803
<20L          4227
```

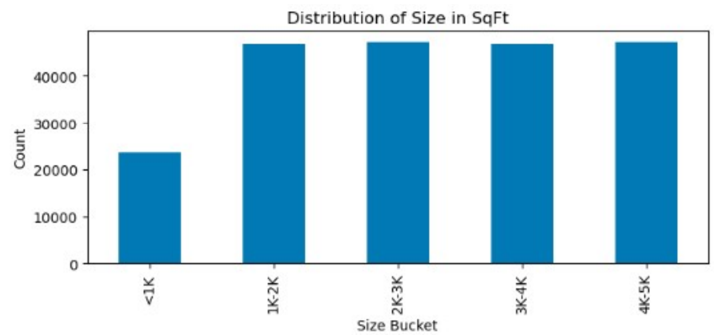


We can see the distribution as uniform distribution but with **48,313 unique values out of ~2.1M rows**, data is **not synthetic uniform data**. Instead, it's likely:
Highly repeated + fairly evenly spread across the range
Majority in **₹2Cr – ₹5Cr range**
Drop after **₹1Cr** - **₹2Cr range**
Very small % in **<₹20L**

What is the distribution of property sizes?




```
size_bucket
2K-3K     47090
4K-5K     47090
3K-4K     46785
1K-2K     46675
<1K       23721
```



The size distribution is essentially **uniform between ~500 and 5000 sqft**, with each bucket having nearly equal counts—no dominant size range.
The smooth KDE and flat bar chart confirm the data is **evenly (likely synthetically) distributed**, not reflecting real-world housing patterns
The counts are nearly uniform across 1K–5K sqft buckets, with only the **<1K range lower due to the truncated starting bound at 500 sqft**

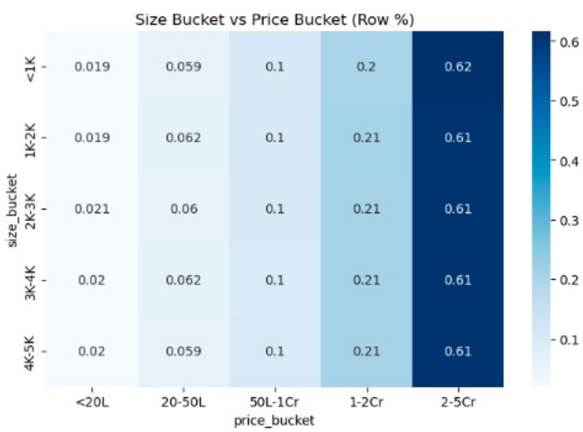
How does price per sq ft vary by property type?

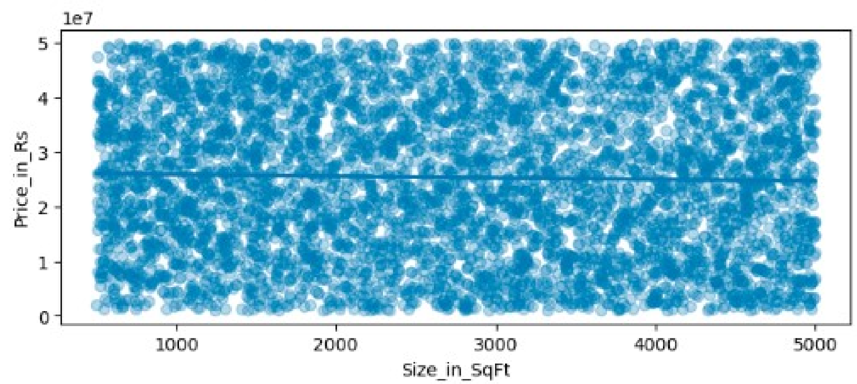
Property_Type	
Apartment	13112.252973
Independent House	13101.549016
Villa	13025.107864

We can **see Price_per_sqft is not affected by the property_type** as the mean is almost ~13000 rupees per_sqft with slightly lower price ~75 rs for Villa

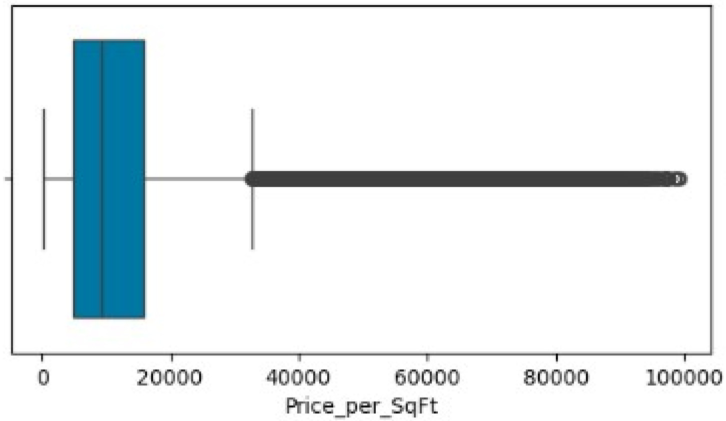
Is there a relationship between property size and price?

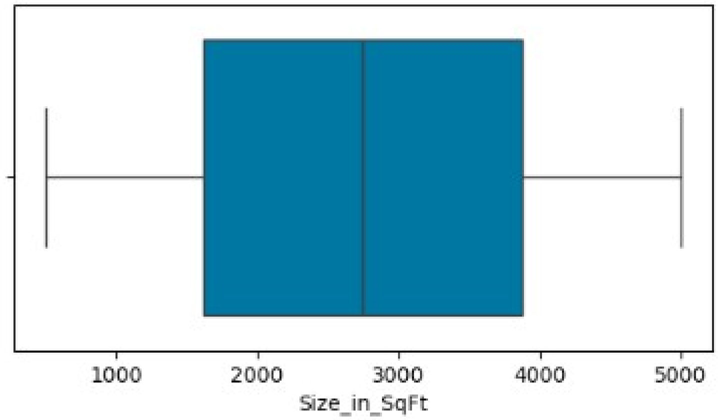
- Price distribution is nearly identical across all size buckets, with ~60% of properties falling in the 2–5Cr range regardless of size.
- This shows **little to no relationship between size and total price**, meaning other factors (or synthetic structure) are dominating pricing.





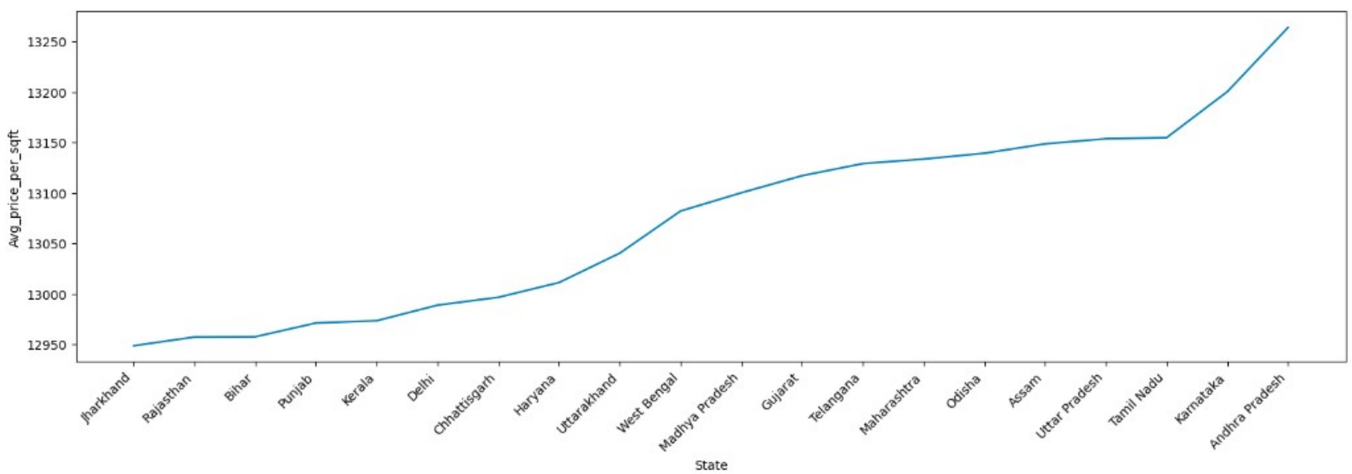
Are there any outliers in price per sq ft or property size?





There are no outliers in Size_per_SqFt but outliers in Price_per_SqFt after ~33000

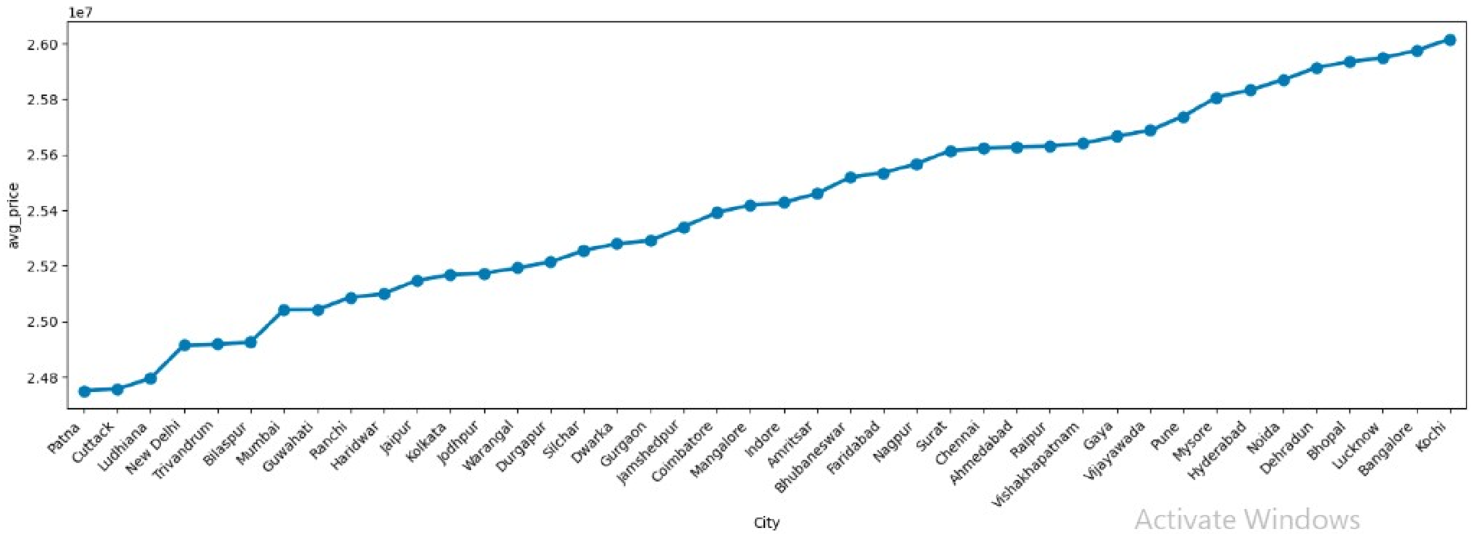
What is the average price per sq ft by state?



State	Avg_price_per_sqft
Jharkhand	12948.617826
Rajasthan	12957.232375
Bihar	12957.510622
Tamil Nadu	13154.681695
Karnataka	13200.274723
Andhra Pradesh	13263.900818

There is no much difference between states average_price_per_sqft as there vary ~300 b/w the top 1 and bottom 1 state

What is the average property price by city?

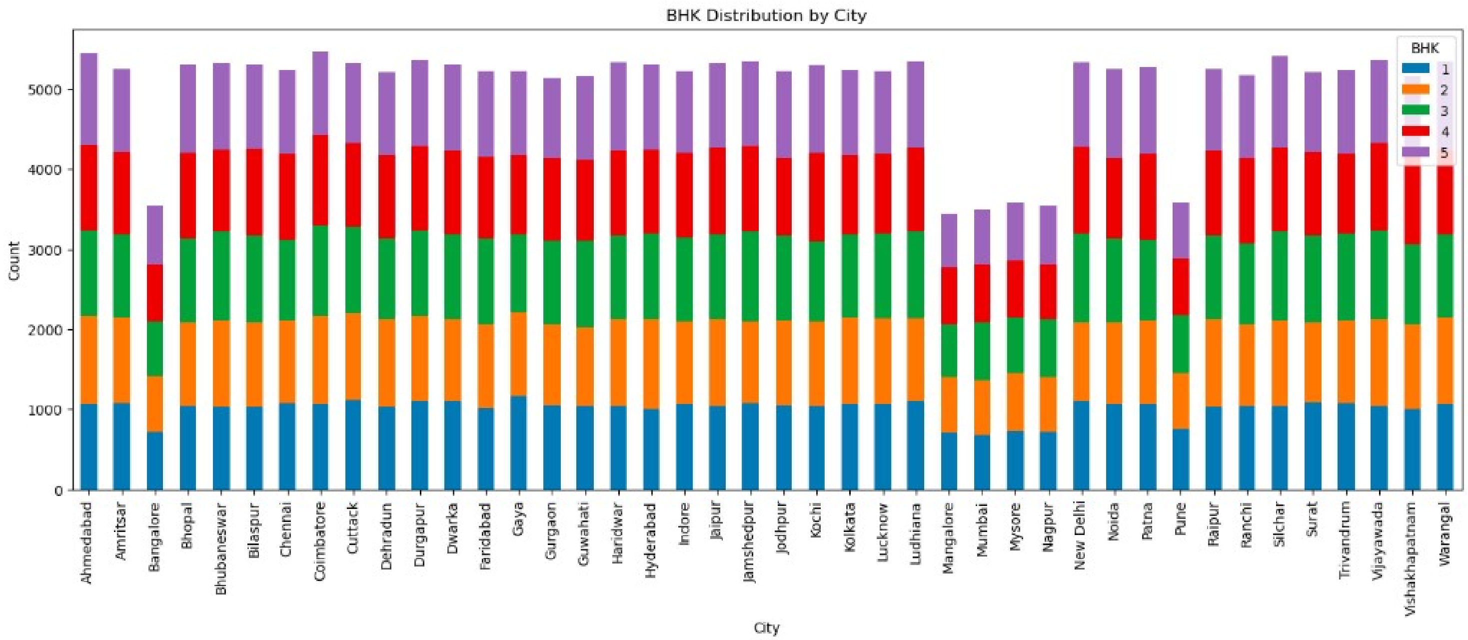


The Average Property price by city ranges between 2.48Cr Patna to 2.60 Cr Kochi (median) . With ~12lakh rupees difference between the highest and lowest price cities

What is the median age of properties by locality?

Median age of Localities vary between 17-22 years

How is BHK distributed across cities?

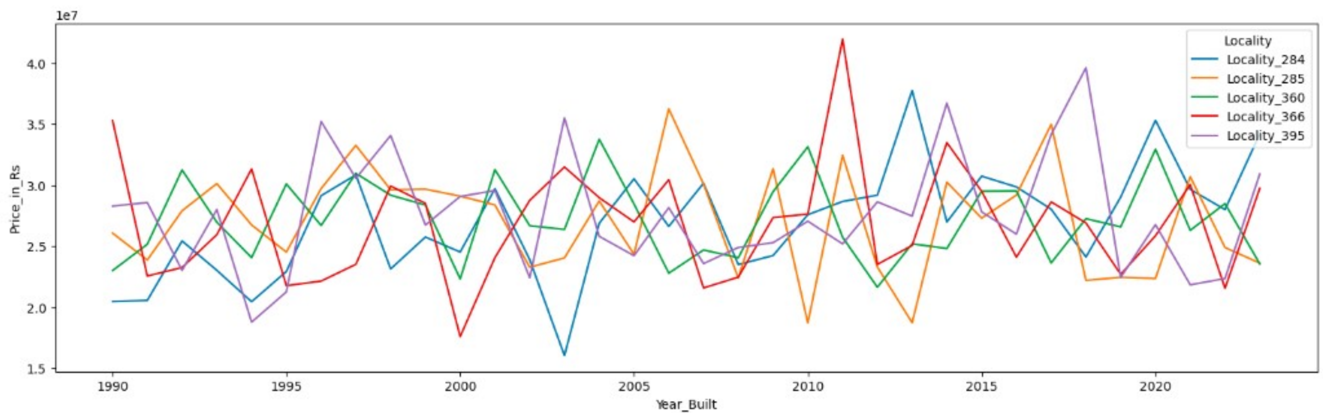


No. of properties
Coimbatore 5476
Ahmedabad 5449
Silchar 5405
Vijayawada 5360
Durgapur 5355
Warangal 5350

Mysore 3580
Pune 3579
Bangalore 3547
Nagpur 3538
Mumbai 3496
Mangalore 3440

We can see BHK are distributed equally among cities. Cannot find much difference in distribution

What are the price trends for the top 5 most expensive localities?

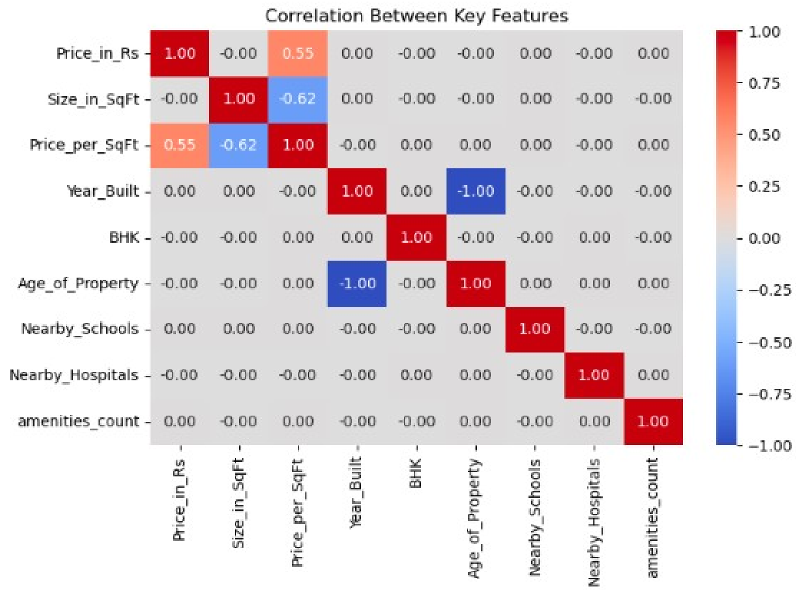


'Locality_395',
'Locality_366',
'Locality_285',
'Locality_284',
'Locality_360'

Locality_284 touched the lower peak of Rs1.60 Cr in 2003

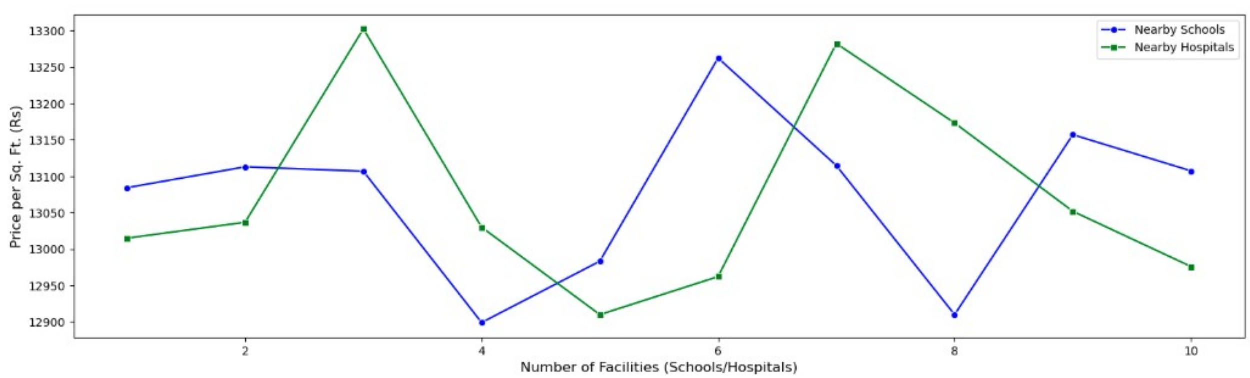
Locality_366 touched the highest peak of Rs4.19 Cr in 2011

How are numeric features correlated with each other?



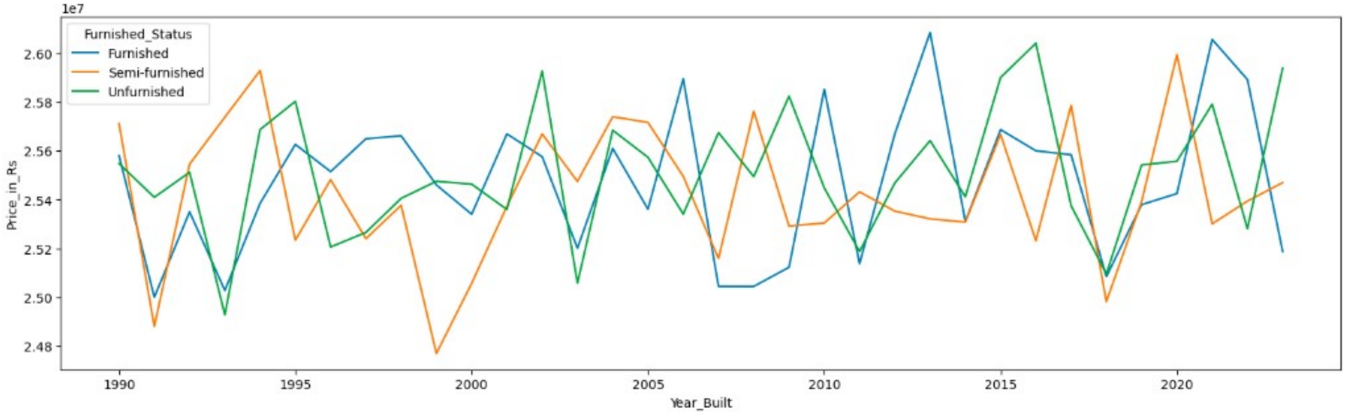
1. **Price is moderately driven by price per square foot**
   The correlation between *Price_in_Rs* and *Price_per_SqFt* (~0.55) suggests that higher per-unit pricing significantly influences overall property price.
2. **Larger homes tend to have lower price per square foot**
   The negative correlation between *Size_in_SqFt* and *Price_per_SqFt* (~ -0.62) indicates a common trend: bigger properties often get a lower per-unit rate.
3. **Age and year built are perfectly inversely related**
   The strong negative correlation (~ -1.00) between *Year_Built* and *Age_of_Property* is expected—newer properties naturally have lower age, confirming data consistency.

How do nearby schools and nearby schools relate to price per sq ft?



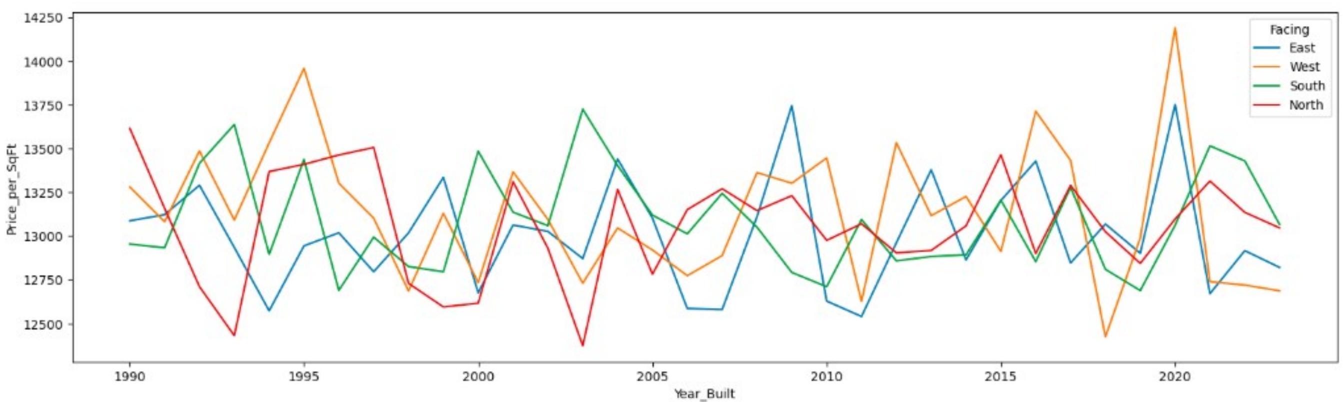
1. More nearby schools generally correlate with slightly higher property prices.
2. Hospital proximity shows sharper price spikes at certain points.
3. Schools have a steadier impact, while hospitals create more variable pricing trends.

How does price vary by furnished status?



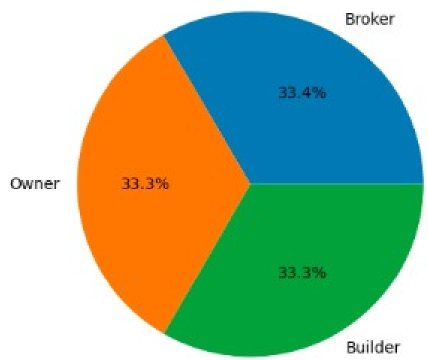
1. Prices are fairly similar across furnishing types.
2. Slight peaks for furnished homes in recent years. Furnished properties occasionally show higher price spikes, especially in newer constructions.
3. No clear upward or downward trend over time

How does price per sq ft vary by property facing direction?



1. **West-facing homes often reach the highest prices**
2. **All facing types fluctuate similarly over time**
3. **Price differences between directions are relatively small**
   While there are occasional spikes, overall the gap between facing types is not very large, indicating direction has limited impact on price per sq ft.

How many properties belong to each owner type?




```
Owner_Type
Broker     70607
Owner      70383
Builder    70371
```

How many properties are available under each availability status?

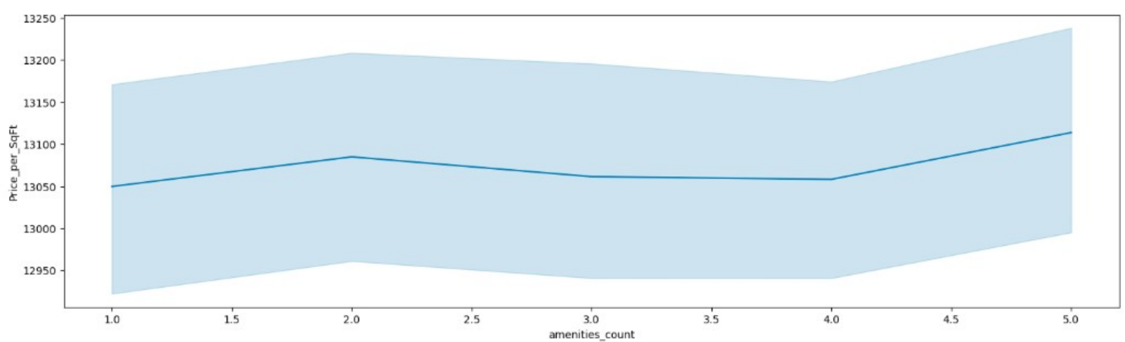
50% are Under_construction and 50% Ready_to_move

Does parking space affect property price?

Difference of ~55k rupees is there between those who have parking space and those who have not

```
amenities_count
1    13049.687413
2    13084.859790
3    13061.366000
4    13058.150782
5    13113.737057
```

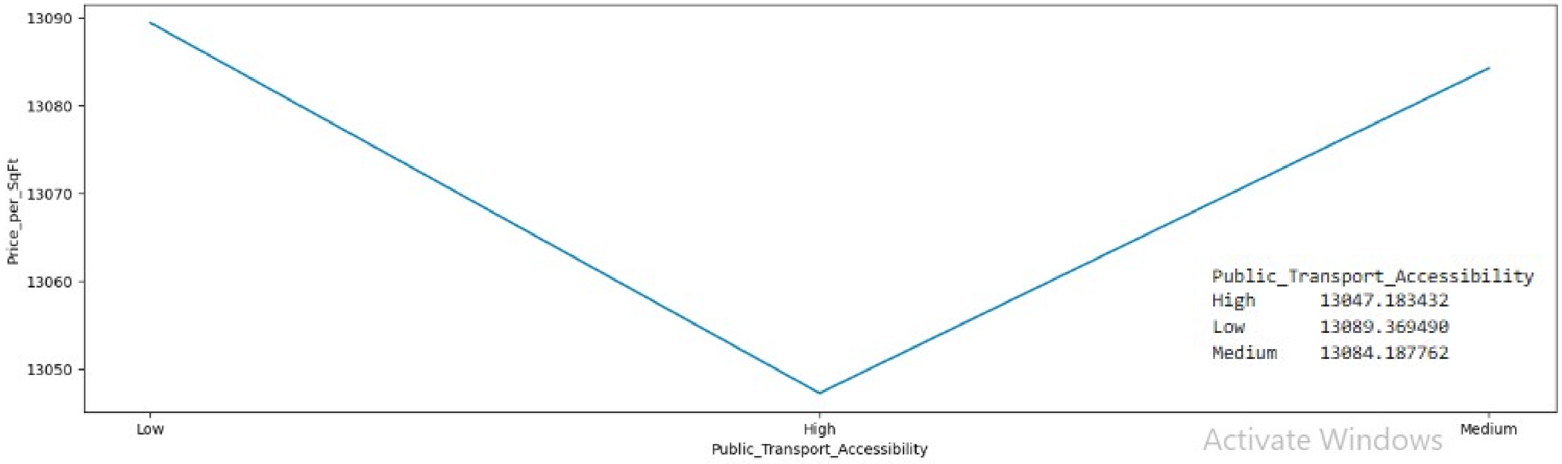
How do amenities affect price per sq ft?



Mild positive relationship: As amenities_count increases, price_per_sqft trends slightly upward. **Small dip in the middle:** There's a noticeable flattening or slight decline around 3–4 amenities, indicating the relationship isn't strictly linear and may be influenced by other factors. **Wide variability:** The shaded band (likely variability or confidence interval) is fairly broad across all points, meaning prices vary significantly even within the same amenities count.

How does public transport accessibility relate to price per sq ft or investment potential?

Price_per_sqft is low for High Transport Accessibility compared to low and medium ~Rs.40 per sq.ft



Investment Insights (1/2)

1. **Larger properties** tend to have lower price per sq.ft, shown by the strong negative correlation between size and price_per_sqft (~ -0.62).
2. **Ready-to-move** properties cost about ₹70,000 more on average than under-construction properties.
3. Properties with **more amenities** generally have higher prices, with around ₹48,000 difference between 1 and 5 amenities.
4. **Owner-listed properties** are cheaper by roughly ₹62,000 compared to broker or builder-listed properties.
5. **Parking availability** increases property value, with about ₹55,000 higher prices for properties having parking space
6. **Security** : Properties with security facilities have an average property value higher by approximately ₹1.46 lakh compared to properties without security.
7. **Cities**: Average property prices across cities range from approximately **₹2.48 Cr in Patna** to **₹2.60 Cr in Kochi**, showing a relatively narrow variation of about **₹12 lakh** overall.
   - **Southern metro cities** like Kochi, Bangalore, Hyderabad, and Chennai appear among the highest-priced markets, indicating stronger demand and urban growth.
   - **Patna, Cuttack, and Ludhiana** show the lowest average prices, making them comparatively more affordable investment locations.
8. **West-facing homes often reach the highest prices** No clear upward or downward trend over time
9. **BHK is not a strong value differentiator**
   Since prices (₹2.539–₹2.551 Cr) and price per sq.ft (~₹13,000) are almost the same across 1–5 BHK, investing based only on BHK won't give much advantage.
   - 1–2 BHK units can be better for rental yield
   - 3 BHK may offer better resale liquidity
   - No "premium pricing" for larger BHKs

Investment Insights (2/2)

10. **States**: Price variation across states is small (~₹4 lakh range)
    - **Price per sq.ft is tightly clustered (~₹12,950–₹13,260)**
      Even the highest (Andhra Pradesh ~₹13,263) and lowest (Jharkhand ~₹12,948) differ by only ~₹300–₹350, indicating **weak state-level pricing impact**.
    - **Investment insight: location within state matters more than state itself**
      Since differences are minimal, returns are likely driven more by **city, micro-location, amenities, and demand**, not just the state.
11. **Furnished_Status**: Unfurnished properties have the highest average price (~₹2.551 Cr)
    - Furnished properties are slightly cheaper (~₹2.547 Cr)
    - **Price variation is very small (~₹8–9 lakh range)**
      This shows furnishing status has **low impact on investment returns**
    - **Rental income → Semi-furnished (higher per sq.ft efficiency) Long-term resale → Unfurnished (slightly higher total value & flexibility)**